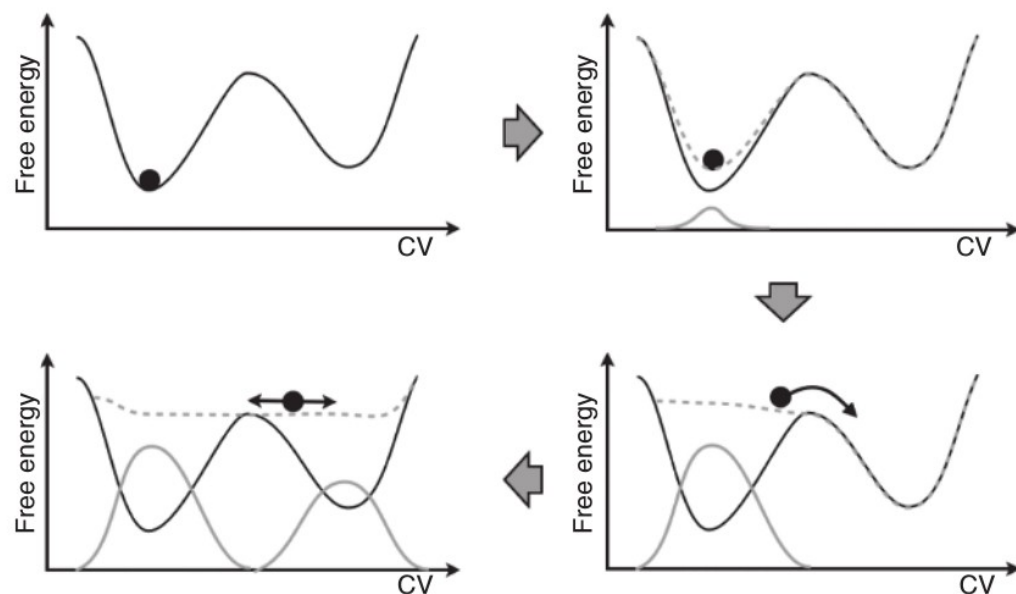


Capabilities of metadynamics

Naf Guo
2026

Metadynamics

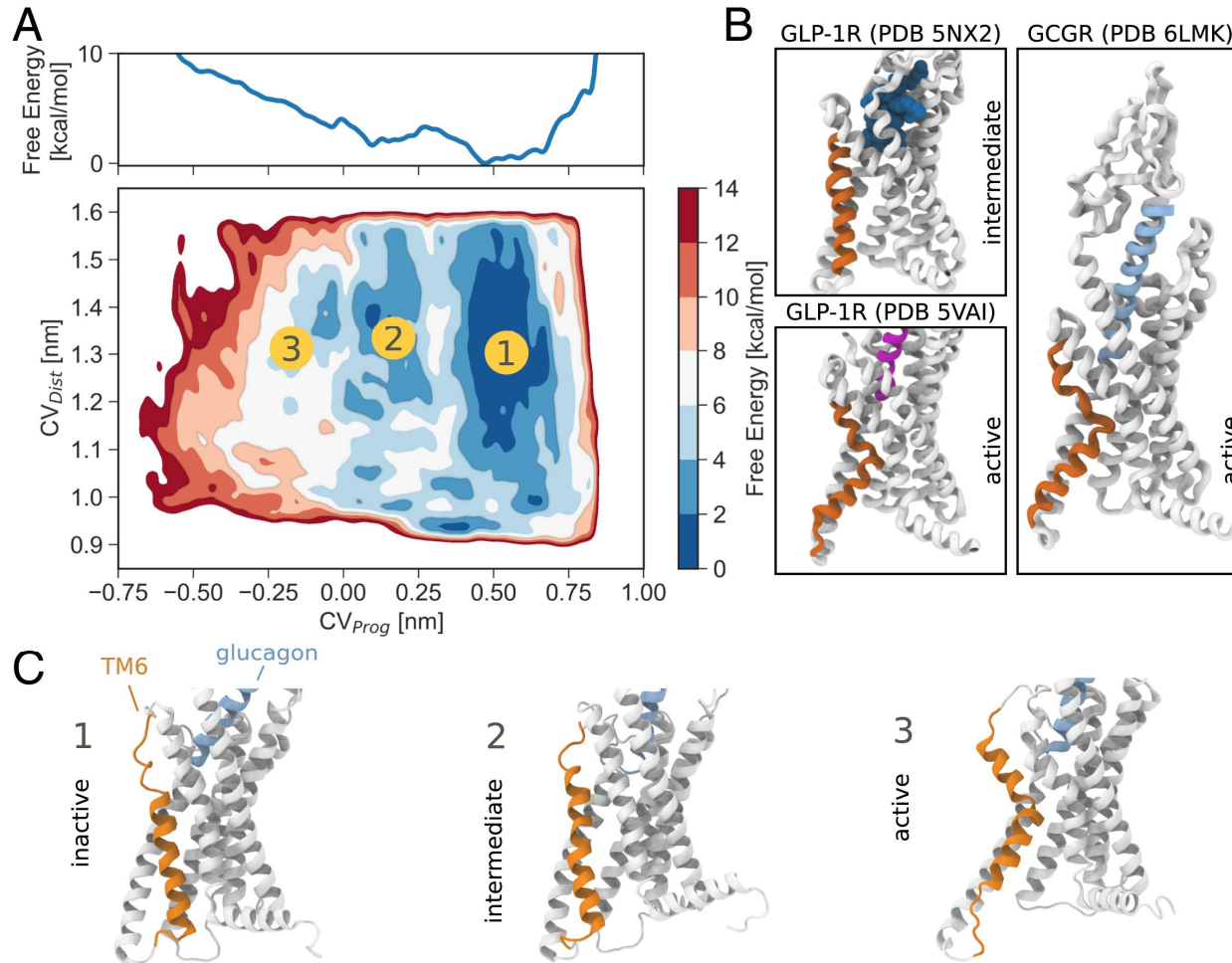
By using only minimal a priori knowledge of the ongoing processes, this method enables one to investigate reaction mechanisms and to compute free energy landscapes. As Parrinello put it in reply to a somewhat nasty question at a conference: "I now moved at least from Newton to Gibbs - from the 17th to the 19th century ...".



A sketch of the process of metadynamics.

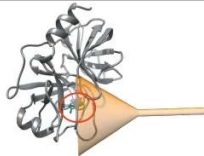
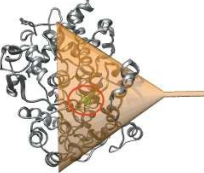
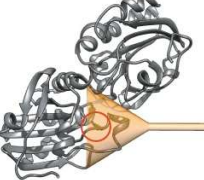
First the system evolves according to a normal dynamics, then a Gaussian potential is deposited (solid gray line). This lifts the system and modifies the free-energy landscape (dashed gray line) in which the dynamics evolves. After a while, the sum of Gaussian potentials fills up the first metastable state and the system moves into the second metastable basin. After this the second metastable basin is filled, at this point, the system evolves in a flat landscape. The summation of the deposited bias (solid gray profile) provides a first rough negative estimate of the free-energy profile.

Simulate the activation of GPCR

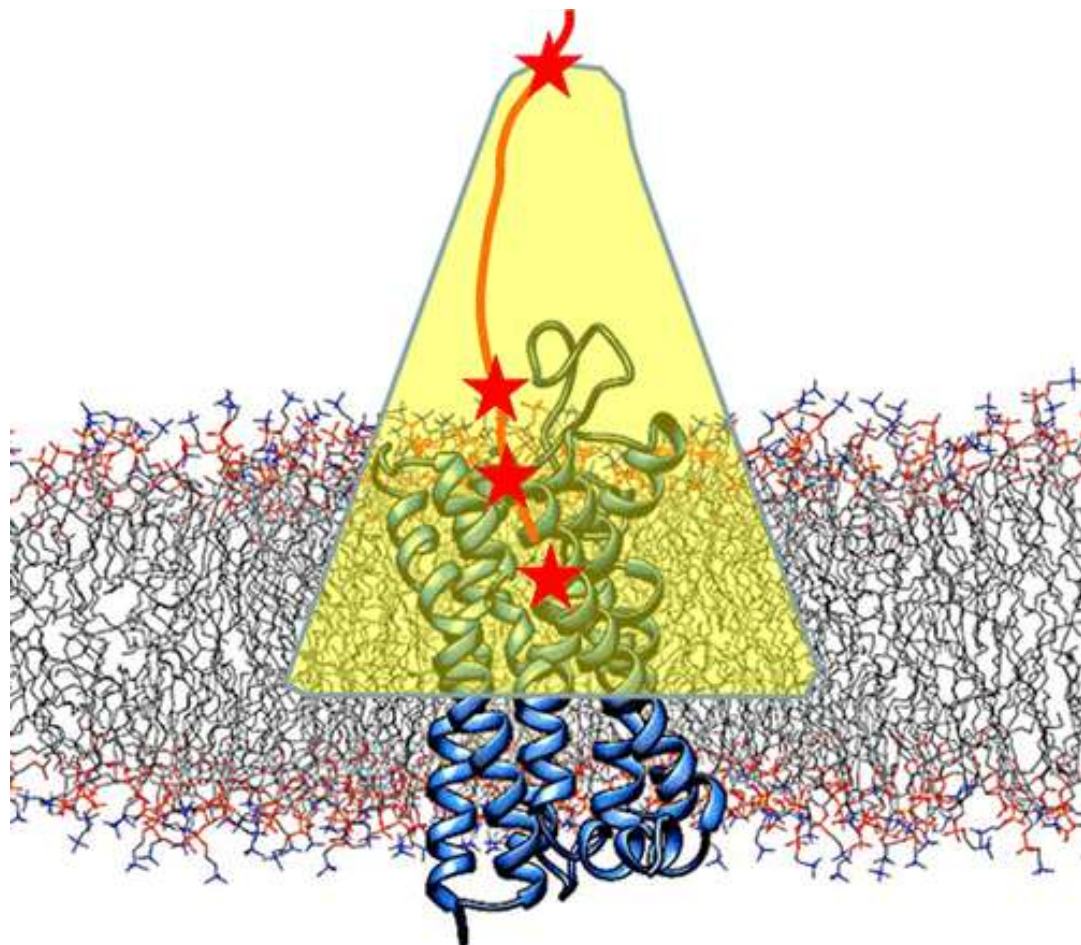


- CV_{prog} was calculated as the difference between the RMSD to the active and inactive structures.
- CV_{Dist} was calculated as the sum between the two values.
- CV_{prog} approximates the reaction coordinate: As it decreases, the receptor transitions from inactive-like to active-like conformations.
- CV_{Dist} is a measure of how far the system deviates from a linear interpolation between inactive and active states, allowing the exploration different activation pathways.
- With these 2 CVs, the simulation can model the activation process of GLP-1R.

Funnel metadynamics for ligand binding free-energy calculation

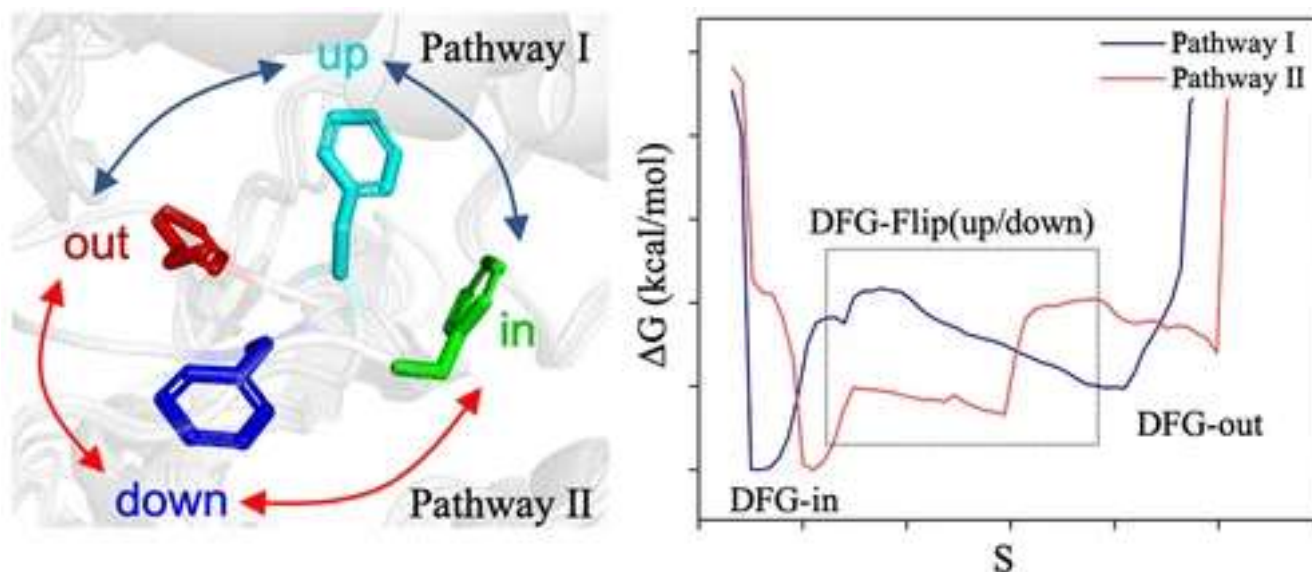
System (atoms)	CVs	Ref.	Image	Zcc (nm)	Alpha (rad)	Rcyl (nm)	Conv. time (ns)	ΔG_{exp} (kcal/mol)	ΔG_{calc} (kcal/mol)
Benzamidine Trypsin 104906	Distance	/		1.8	0.55	0.1	250	-7.2 to -6.3	-8.2 ± 0.5
	Distance								
	Torsion								
SC-558 COX-2 57208	Distance	5		4.4	0.6	0.1	350	-12	-11.1 ± 1.5
	Torsion								
	C-map								
Berberine G4s 28093	Distance	11		1.6	0.75	0.1	450	-9.9 to -9.3	-10.3 ± 0.5
	Torsion								
Catechol/ 4-me-catechol peroxiredoxin 5 99583	Distance	9		1.8	0.55	0.1	300-350	-3.2 ± 0.1 -4.1 ± 0.1	-3.0 ± 0.2 -4.2 ± 0.2

Funnel metadynamics calculate ligand binding free-energy for GPCR



Saleh, Nouredin, et al. "An efficient metadynamics-based protocol to model the binding affinity and the transition state ensemble of G-protein-coupled receptor ligands." *Journal of chemical information and modeling* 57.5 (2017): 1210-1217.

Metadynamics to model conformational transition between DFG-in and DFG-out



Jiang, Tao, et al. "The conformational transition pathways and hidden intermediates in DFG-flip process of c-Met kinase revealed by metadynamics simulations." *Journal of Chemical Information and Modeling* 62.15 (2022): 3651-3663.

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